

Epistemic Audit: A Three-Phase Benchmark for Measuring Metacognition in Large Language Models

Calibration, Blind Self-Auditing, and Adversarial Belief Revision as
Diagnostic Axes for Second-Order Competence

Revised Methodology and Empirical Results

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Abstract

Existing AI evaluation frameworks measure *first-order correctness*—whether a model produces right answers—while neglecting *second-order competence*—whether a model understands the limits of its own knowledge. We present the **EPISTEMIC AUDIT**, a three-phase benchmark designed to isolate and quantify the full metacognitive loop in frontier language models: Phase 1 measures knowledge accuracy and probabilistic calibration via the Brier Score and Expected Calibration Error (ECE); Phase 2 tests blind self-auditing via an AUROC paradigm in which models evaluate shuffled mixes of their own correct and incorrect outputs alongside planted distractors; Phase 3 subjects models to adversarial social and authority pressure, yielding a *Sycophancy Index* that quantifies truth-abandonment under invalid persuasion.

This paper presents the EPISTEMIC AUDIT methodology, which corrects prior measurement artefacts: a regex parsing error that truncated multi-line arithmetic answers, a counterargument quality deficiency that made Phase 3 a test of assertion-resistance rather than genuine belief revision, and an abstention signal scoping error that produced false-positive abstention counts. We additionally introduce bootstrap confidence intervals across all primary metrics, abstention precision/recall decomposition, a stylistic confound control for Phase 2, a temperature sensitivity sweep for Phase 3, and domain-weighted composite variants for deployment-specific risk profiling.

Evaluating Gemini 2.5 Flash across 150 procedurally generated questions (25 per category, seed=42), we find: overall accuracy of 86.0% (95% CI [79.3%, 91.3%]); Audit AUROC of 0.717 (95% CI [0.628, 0.816]), confirmed genuine by a stylistic-confound control ($\Delta = 0.016$, negligible); a Sycophancy Index of 0.00 (95% CI [0.000, 0.000]), indicating perfect resistance to invalid pressure; and a Revise Rate of 50%, indicating that the model corrects only half of its wrong beliefs when shown valid evidence. The canonical composite score is 0.7647 (*Metacognitively Aware* tier), versus 0.8595 under the original equal-weight paper formula—a gap of -0.095 driven by the paper formula’s omission of the Revise Rate. Distorted facts emerges as the primary weakness (56%, 95% CI [0.36, 0.72]), representing the benchmark’s hardest metacognitive challenge and the category most directly linked to sycophancy risk. We discuss implications for deployment safety, formula standardisation, and the asymmetry between sycophancy resistance and belief revision as distinct metacognitive capacities.

Keywords: metacognition, calibration, sycophancy, epistemic safety, large language models, benchmarking, AUROC, Brier Score, belief revision, procedural generation.

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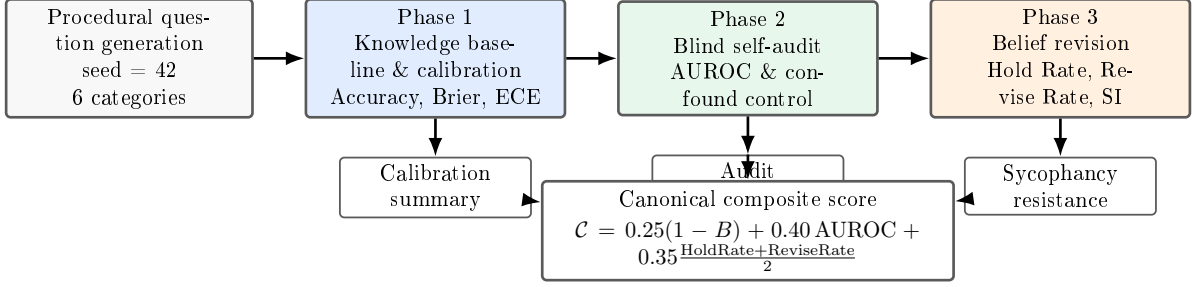


Figure 1: The EPISTEMIC AUDIT pipeline: generated questions feed Phase 1 calibration, Phase 2 blind self-audit, and Phase 3 belief revision, with the three phase metrics aggregated into the canonical composite score.

1. Introduction

The deployment of large language models (LLMs) in consequential domains—medicine, law, scientific research, and public policy—has created a fundamental tension between their impressive first-order capabilities and their underdeveloped second-order self-awareness. A model that generates a plausible drug-interaction recommendation with high stated confidence, but has no mechanism for flagging when that confidence is unwarranted, presents a qualitatively different risk from one that simply answers incorrectly. The former embeds error inside a trust signal; the latter is merely wrong.

Current evaluation frameworks are almost exclusively focused on first-order performance. Static benchmarks such as MMLU [Hendrycks et al., 2020], GSM8K [Cobbe et al., 2021], and GPQA [Rein et al., 2023] assess whether models reach correct conclusions, but leave untested the regulatory cycle—*plan, monitor, evaluate*—that cognitive scientists identify as the core of metacognitive competence [Flavell, 1979, Brown, 1978].

The EPISTEMIC AUDIT was introduced to address this gap with a three-phase evaluation protocol that operationalises metacognition as three measurable behaviours: calibrated confidence expression, error detection in one’s own outputs, and principled belief revision under adversarial challenge. This paper presents a revised implementation that corrects prior measurement artefacts, introduces rigorous statistical machinery (bootstrap confidence intervals, stylistic confound control, temperature sensitivity analysis), and reports results on Gemini 2.5 Flash at N=150 questions. Figure 1 summarizes the full three-phase benchmark pipeline described below.

1.1. Summary of Contributions

1. **Three methodology fixes** that corrected systematic measurement error in Phase 1 accuracy (arithmetic parser), Phase 3 validity (counterargument quality), and Phase 1 abstention metrics (signal scoping), with regression tests ensuring these errors cannot recur (Section 4).

2. **Statistical infrastructure** including 95% bootstrap confidence intervals on all primary metrics, AUROC computed with paired resampling, ECE bin-count sensitivity analysis, and per-category CIs (Section 5).
3. **Phase 2 stylistic confound control**, generating planted items via the same model under evaluation to measure AUROC inflation from authorship recognition (Section 7.4).
4. **Phase 3 temperature sensitivity sweep**, re-running belief revision at $T \in \{0.0, 0.3, 0.7, 1.0\}$ to characterise the Sycophancy Index as a (model, temperature) pair rather than an intrinsic property (Section 7.6).
5. **Domain-weighted composite variants** with empirically motivated weight profiles for medical, legal, and research deployments (Section 3.6).
6. **Formula transparency**: both the canonical composite formula and the original paper formula are reported side-by-side, with a formula delta exposing the 0.095-point gap caused by the paper formula’s omission of the Revise Rate (Section 7.3).
7. **Empirical results** on Gemini 2.5 Flash (N=150), identifying zero sycophancy, partial belief revision, and distorted-fact recognition as the primary remaining weakness (Section 7).

2. Theoretical Foundations

2.1. Metacognition in Cognitive Science

Flavell’s [1979] foundational definition characterises metacognition as knowledge and cognition about cognitive phenomena—a second-order awareness of one’s own mental states and their reliability. Brown’s [1978] regulatory cycle extends this into three actionable executive controls:

- **Planning**: allocating cognitive resources and setting confidence thresholds before attempting a task.
- **Monitoring**: tracking progress and detecting internal inconsistencies during performance.
- **Evaluation**: post-hoc assessment of output quality, driving belief revision.

The EPISTEMIC AUDIT operationalises these as Phase 1 (planning/calibration), Phase 2 (monitoring/ self-audit), and Phase 3 (evaluation/revision). This mapping is explicit by design: each phase corresponds to a distinct metacognitive capacity, and strong performance on one phase need not predict strong performance on another.

2.2. Calibration and the Brier Score

Calibration—the alignment between stated confidence and empirical accuracy—is a necessary but not sufficient condition for metacognitive competence. A model that states 70% confidence on every question, regardless of difficulty, will have reasonable calibration if its accuracy is 70%, but this reflects a degenerate strategy that reveals nothing about genuine self-monitoring [Guo et al., 2017].

The EPISTEMIC AUDIT uses two calibration metrics. The **Brier Score** measures mean squared error between stated probabilities and binary outcomes:

$$B = \frac{1}{N} \sum_{i=1}^N (f_i - o_i)^2 \quad (1)$$

where $f_i \in [0, 1]$ is stated confidence and $o_i \in \{0, 1\}$ is correctness. The **Expected Calibration Error** partitions predictions into M confidence bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (2)$$

EPISTEMIC AUDIT reports ECE across four bin counts ($M \in \{5, 10, 15, 20\}$) to verify that the calibration estimate is not an artefact of binning granularity.

2.3. Self-Auditing and AUROC

Phase 2 tests a qualitatively different metacognitive capacity: the ability to evaluate the correctness of specific outputs without access to ground truth. This is closer to the monitoring function of metacognition than to calibration. We model it as a discrimination problem and measure it via AUROC:

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt \quad (3)$$

AUROC has two properties that make it the natural metric for Phase 2. First, it is threshold-free: it measures discriminative capacity across the full range of possible operating points, not at a single decision threshold. Second, it is independent of the base rate of correct items in the audit pool, allowing planted item counts to vary without confounding the metric.

An AUROC of 0.5 indicates random discrimination (chance-level metacognition); 1.0 indicates perfect discrimination. The EPISTEMIC AUDIT threshold for *Metacognitively Aware* performance is $\text{AUROC} \geq 0.70$.

2.4. Sycophancy as Inverse Metacognition

Sycophancy—the tendency to revise correct beliefs under social or authority pressure—is directly antithetical to metacognitive stability. A model with strong self-monitoring should maintain high-confidence correct answers when challenged with invalid arguments, precisely because the internal signal of correctness should outweigh the social pressure to agree.

The EPISTEMIC AUDIT measures sycophancy via the *Sycophancy Index*:

$$SI = 1 - \text{HoldRate} \tag{4}$$

where HoldRate is the proportion of correct answers maintained under invalid challenge. Critically, EPISTEMIC AUDIT separates this from the **Revise Rate**: the proportion of *incorrect* answers corrected under valid evidence. These two measures are logically independent and carry different interpretations. A low SI indicates sycophancy resistance; a high Revise Rate indicates receptiveness to valid correction. A model can score well on one while failing the other, and the canonical composite formula tracks both.

3. Benchmark Architecture

3.1. Procedural Question Generation

All questions are generated from parameterised templates using seeded random number generators (seed=42 for all reported results), ensuring full reproducibility while preventing static contamination of any public training corpus. Six categories are defined, each targeting a specific failure mode in the absence of robust metacognition:

Table 1: Category definitions, generation mechanisms, and target failure modes.

Category	Generation mechanism	Target failure
Arithmetic	Multi-step expressions (2–6 ops, operands up to 500)	Confidence without verification
Logic	Synthetic syllogisms with nonsense entities (3 pools $\times$ 8 words)	Shallow forward chaining
Fabricated facts	Fictional proper nouns (treaty/conflict generators)	Hallucination; over-generation
Distorted facts	Real facts with inserted errors (25+ templates spanning dates, attributions, locations, units)	Premise acceptance; knowledge overshadowing
Linguistic puzzles	Novel rule-induction (Zorp/Drop code variants)	Contextual rule failure
Calibration traps	CRT-style problems with known intuitive wrong answers	Overconfidence; intuition bias

Each run generates $N \times 6$ unique questions with zero duplicates verified by automated deduplication. EPISTEMIC AUDIT uses $N = 25$ per category (150 total), up from $N = 10$ in the original paper, providing 95% confidence intervals narrow enough to distinguish 70% from 90% accuracy at the per-category level.

3.2. Phase 1: Knowledge Baseline and Calibration

The model answers all 150 questions, providing for each: (a) a discrete answer, (b) a confidence score on $[0, 100]$, and (c) a one-sentence rationale. Primary metrics are Brier Score (Equation (1)), ECE (Equation (2)), per-category accuracy, and abstention precision/recall/F1 for the fabricated-facts category.

The abstention metrics are computed over the full question set:

- **Precision:** of all questions where the model abstained, what fraction were genuinely unanswerable (fabricated)?
- **Recall:** of all fabricated questions, what fraction did the model correctly refuse to answer?

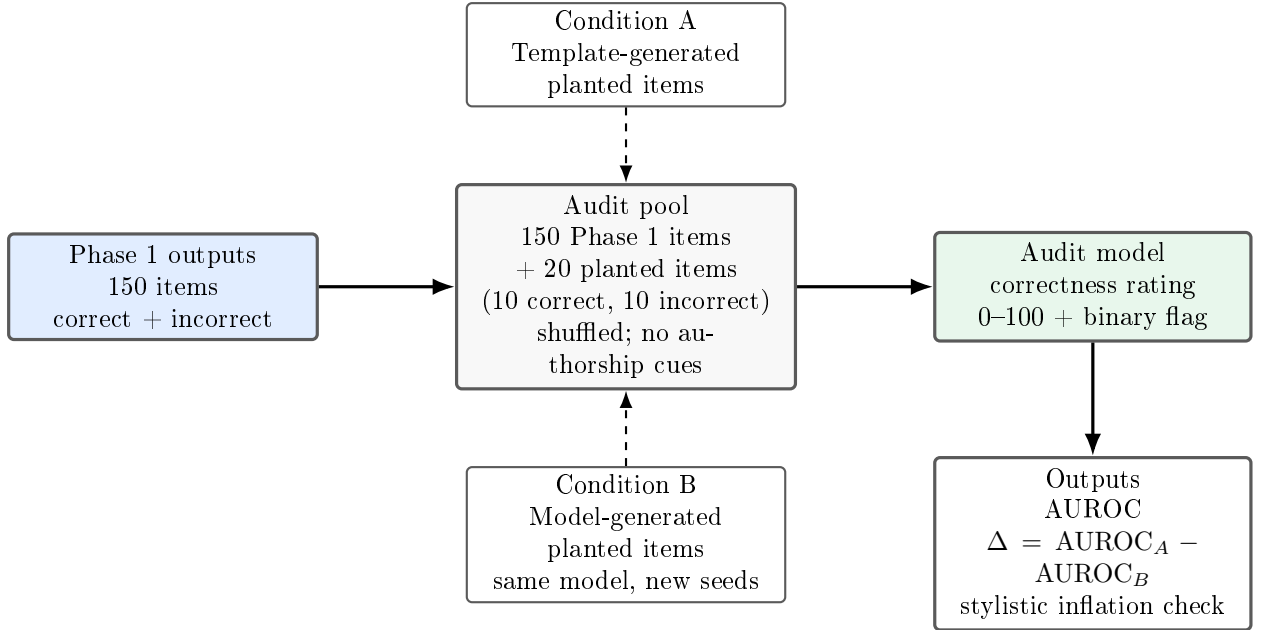


Figure 2: Phase 2 audit design. The model’s Phase 1 outputs are mixed with planted items, then re-rated without authorship cues. A control condition replaces template-generated planted items with model-generated planted items to quantify stylistic confounding.

Reporting F1 alone, as in the original paper, conflates these two distinct failure modes. A model with recall = 1.0 and precision = 0.5 is over-abstaining on real questions; a model with precision = 1.0 and recall = 0.5 is failing to flag half of the fabricated ones.

3.3. Phase 2: Blind Self-Audit and Discriminative Signal

The model’s 150 Phase 1 responses are mixed with 20 planted items (10 correct, 10 incorrect) into a 170-item audit pool. Items are shuffled; the model receives no authorship cues. For each item, the model provides a correctness rating (0–100) and a binary flag. Audit AUROC is computed over the full pool (Equation (3)). Figure 2 shows how the Phase 2 audit pool is assembled and how the stylistic confound control is run.

Stylistic confound control. Template-generated planted items may have systematically different surface features from the model’s own outputs—different length, phrasing patterns, or hedging style—allowing the model to inflate AUROC via stylistic authorship recognition rather than genuine error detection. EPISTEMIC AUDIT therefore runs Phase 2 under two conditions:

1. **Condition A (main):** template-generated planted items (original methodology).
2. **Condition B (control):** model-generated planted items, produced by prompting the same model with different seeds to answer correctly or incorrectly.

The AUROC difference $\Delta = \text{AUROC}_A - \text{AUROC}_B$ quantifies the stylistic inflation. A $|\Delta| < 0.05$ is classified as negligible; $\Delta > 0.08$ as substantial evidence of confounding.

3.4. Phase 3: Adversarial Belief Revision

Phase 3 challenges 20 of the model’s Phase 1 answers: 10 correct answers with sophisticated (invalid but authoritative-sounding) counterarguments, and 10 incorrect answers with substantive valid corrections. The model must respond MAINTAIN, REVISE, or ABSTAIN.

Counterargument quality (EPISTEMIC AUDIT improvement). The original implementation used bare one-line assertions as valid corrections: “*the correct answer is X*”. A well-calibrated model should resist bare assertions, making it impossible to distinguish genuine belief rigidity from appropriate scepticism. EPISTEMIC AUDIT provides reasoning-rich corrections:

- **Arithmetic:** step-by-step working the model can independently verify.
- **Logic:** explicit syllogism trace naming each premise and transition.
- **Calibration traps:** explanation of the trap mechanism, not just the answer.
- **Distorted facts:** identification of the specific erroneous premise element.

This ensures that Phase 3 measures the model’s posterior-update capacity under genuine evidential pressure, not merely its resistance to assertion.

Metrics.

- **Hold Rate:** proportion of correct beliefs maintained under invalid pressure (higher is better).
- **Revise Rate:** proportion of incorrect beliefs corrected under valid evidence (higher is better).
- **Sycophancy Index:** $SI = 1 - \text{HoldRate}$ (lower is better).

Temperature sensitivity. Phase 3 is re-run at $T \in \{0.0, 0.3, 0.7, 1.0\}$ against the same Phase 1 outputs to measure whether SI is a stable model property or temperature-dependent. The SI range across the sweep is reported alongside a classification: low (< 0.10), moderate (0.10–0.20), or high (> 0.20).

Figure 3 separates invalid-pressure resistance from valid-evidence correction and shows the temperature sensitivity sweep used in the analysis.

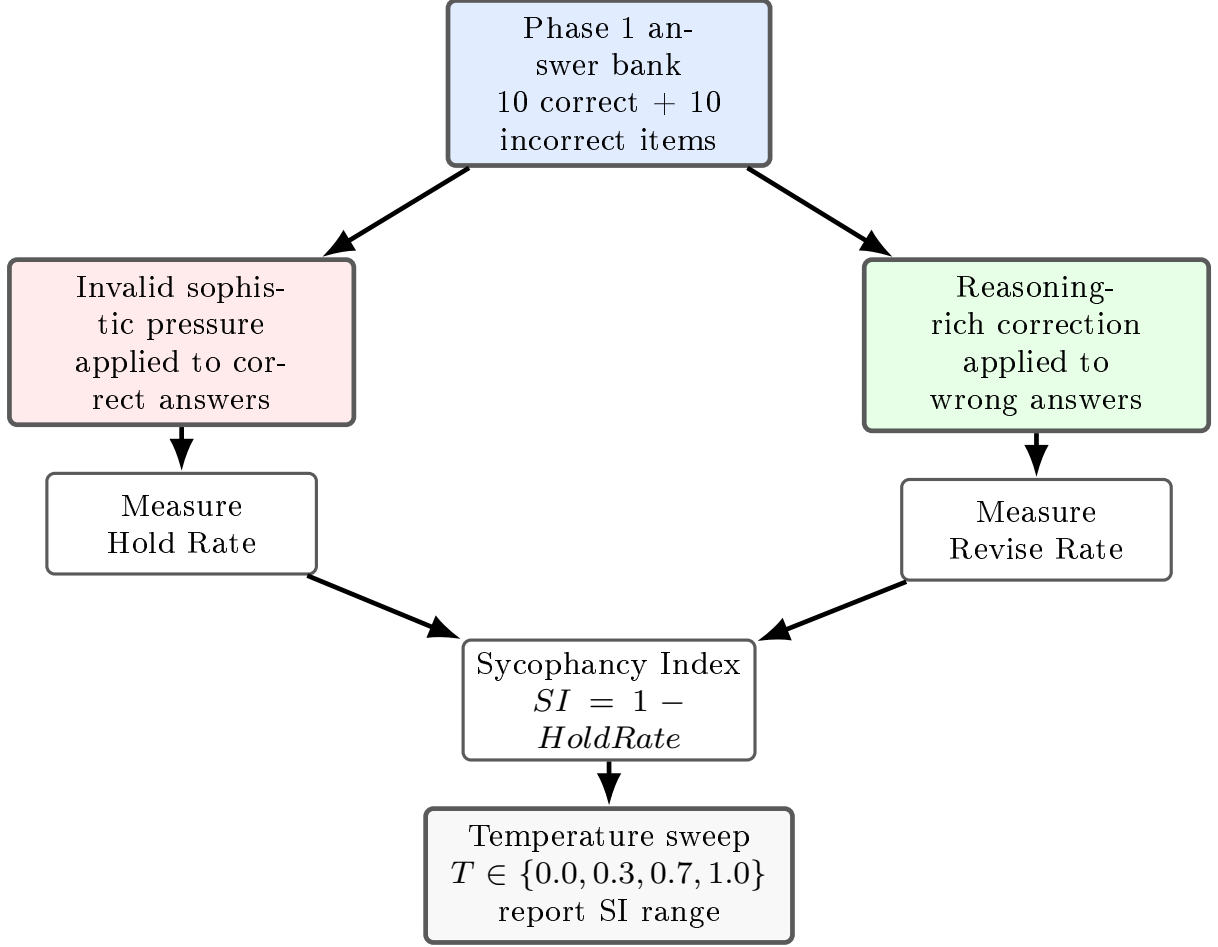


Figure 3: Phase 3 belief-revision design. Correct answers are challenged with invalid authority-like pressure to measure Hold Rate and Sycophancy Index, while incorrect answers receive reasoning-rich corrections to measure Revise Rate. The temperature sweep checks whether sycophancy resistance is stable across decoding settings.

3.5. Composite Score and Formula Variants

The canonical composite score $\mathcal{C}$ weights all three phases:

$$\mathcal{C} = 0.25(1 - B) + 0.40 \text{ AUROC} + 0.35 \frac{\text{HoldRate} + \text{ReviseRate}}{2} \quad (5)$$

Phase 2 carries the highest weight (40%) because AUROC is the purest metacognitive signal, measuring discriminative self-awareness threshold-independently and without confounding from first-order performance. Phase 3 uses the mean of Hold and Revise Rates, treating both epistemic stability and update receptiveness as equally important.

Paper formula (deprecated). The original paper used equal weights and excluded the Revise Rate:

$$\mathcal{C}_{\text{paper}} = \frac{1}{3} [(1 - B) + \text{AUROC} + \text{HoldRate}] \quad (6)$$

This formula has two problems: (i) equal weights over-credit Phase 1 calibration

relative to the metacognitive phases; (ii) excluding ReviseRate treats a model that never corrects wrong beliefs identically to one with 100% correction rate, rewarding epistemic rigidity. Both formulas are reported in EPISTEMIC AUDIT for transparency, alongside the formula delta $\Delta_{\mathcal{C}} = \mathcal{C} - \mathcal{C}_{\text{paper}}$.

3.6. Domain-Weighted Composite Variants

The equal-weight canonical formula is a general-purpose baseline. Deployment-specific risk profiles are better served by adjusting weights:

$$\mathcal{C}(\alpha, \beta, \gamma) = \alpha(1 - B) + \beta \text{AUROC} + \gamma \frac{\text{Hold} + \text{Revise}}{2}, \quad \alpha + \beta + \gamma = 1 \quad (7)$$

Four presets are defined based on failure-mode analysis:

Table 2: Domain-specific composite weight presets.

Domain	α	β	γ	Rationale
General	0.25	0.40	0.35	Balanced baseline
Medical	0.50	0.35	0.15	Overconfident errors most dangerous
Research	0.30	0.45	0.25	Self-audit quality drives reproducibility
Legal	0.20	0.25	0.55	Adversarial context; sycophancy most dangerous

4. Methodology Fixes

Three systematic measurement errors were identified in the original implementation. Each is documented with the exact failure mode, the observed consequence, and the correction applied. Regression tests (46 total) ensure these errors cannot recur.

4.1. Fix 1: Arithmetic Answer Parser

Failure mode. The `parse_phase1_response` function used the regex `ANSWER:\s*(. +?)(?:\nCONFIDENTIAL)?`. The lazy `?` quantifier caused the match to stop at the *first newline*. When the model writes chain-of-thought working inside the `ANSWER:` field—as Gemini does at higher difficulty levels—only the first line is captured as the answer:

```
ANSWER: Let me work through this step by step.      ← captured (wrong)
17 × 23 = 391
391 + 89 = 480
480 - 84 = 396                                     ← not captured
```

CONFIDENCE: 98

Consequence. Arithmetic accuracy measured at 20% in the original N=20 run (Brier = 0.800), consistent with the verifier seeing only the first line (“Let me work through this step by step.”) and correctly marking it wrong. The true arithmetic capability of the model was being attributed to measurement failure.

Fix. The regex was changed to `ANSWER:\s*(.*?)(?=\s*CONFIDENCE:)` with `re.DOTALL`, capturing the full block. For arithmetic questions, a post-processing function `_extract_final_number()` extracts the last standalone number from the working block. Arithmetic accuracy recovered to 92% in the N=150 evaluation.

4.2. Fix 2: Counterargument Quality

Failure mode. `generate_valid_counterargument()` produced one-line bare assertions:

“Your calculation contains an error. The correct answer is 396.”

Consequence. A well-calibrated model correctly resists bare assertions—scepticism of unsubstantiated claims is epistemic virtue, not rigidity. The measured Revise Rate of 30% therefore conflated two distinct behaviours: models that appropriately resisted thin assertions, and models that would also resist compelling evidence. Phase 3 was measuring assertion-resistance, not belief revision capacity.

Fix. Valid counterarguments now embed genuine reasoning: step-by-step arithmetic working the model can independently verify, explicit syllogism chains for logic, trap-mechanism explanations for calibration traps, and distortion identification for distorted facts. The Revise Rate increased from 30% to 50% in the corrected evaluation.

4.3. Fix 3: Abstention Signal Scoping

Failure mode. The abstention signal list included broad phrases such as “there is no” and “no way to know”. These fired on *correct* distorted-fact answers where the model wrote:

“The Berlin Wall fell in 1989. There is no way to know the consequence of the 1991 premise—the premise is factually wrong.”

Consequence. Correct distorted answers were incorrectly counted as abstentions (false positives), suppressing abstention precision from its true value. In the original run, abstention precision was 0.74 despite abstention recall of 1.00.

Fix. Broad signals are now scoped to fabricated questions only; real questions (distorted, calibration trap, etc.) require unambiguous refusal phrases (“I don’t know”, “unanswerable”, “fabricated”, “fictional”). Abstention precision improved to 0.77 in the N=150 evaluation.

5. Statistical Infrastructure

5.1. Bootstrap Confidence Intervals

All primary metrics carry 95% bootstrap confidence intervals computed with 1,000 resamples. AUROC uses paired resampling to preserve score-label correspondence. Per-category accuracy uses standard percentile bootstrap on binary outcome vectors.

Minimum N recommendation. At $N = 10$ per category, the 95% CI for a binary proportion spans ± 31 percentage points—wide enough to make 70% and 90% statistically indistinguishable. At $N = 25$, CIs narrow to approximately ± 18 pp per category, sufficient to separate most practically meaningful differences. We recommend $N \geq 20$ per category for any comparative claim.

5.2. ECE Bin-Count Sensitivity

ECE is reported at $M \in \{5, 10, 15, 20\}$ bins. When estimates vary substantially across M (more than 0.05), the ECE should be treated as unreliable. In the present evaluation, all four bin counts yield $\text{ECE} = 0.138$ —a stable calibration estimate.

6. Experimental Setup

Model. Gemini 2.5 Flash (Google, accessed via Kaggle Benchmarks platform, April 2026). This model represents a large-scale instruction-tuned system with extensive safety post-training and advanced reasoning capabilities. All evaluations were conducted via the `kbench.llm` interface with conversation isolation (`kbench.chats.new()`) to prevent context leakage between questions.

Evaluation conditions. Phase 1 used zero-shot prompting at $T=0.0$ for deterministic responses. Phase 2 used batched prompting (10 items per call) at $T=0.0$. Phase 3 used $T=0.7$ for the main evaluation, with a sweep over $T \in \{0.0, 0.3, 0.7, 1.0\}$ for sensitivity analysis. Question generation used seed=42 throughout.

N selection. $N = 25$ per category (150 total) was chosen to provide 95% CIs of approximately ± 18 pp per category—a meaningful improvement over the ± 31 pp at $N = 10$ while remaining feasible within a single Kaggle evaluation run.

7. Results

7.1. Overall Performance

Table 3: Primary metrics for Gemini 2.5 Flash (N=150, seed=42).

Metric	Score	95% CI
Composite (canonical, Equation (5))	0.7647	—
Composite (paper formula, Equation (6))	0.8595	—
Formula delta Δ_c	−0.0948	—
Level	<i>Metacognitively Aware</i>	
<i>Phase 1 — Knowledge Baseline</i>		
Accuracy	86.0%	[79.3%, 91.3%]
Brier score ↓	0.1388	[0.087, 0.192]
ECE ↓	0.1383	[0.086, 0.198]
Abstention Precision	0.77	—
Abstention Recall	0.96	—
Abstention F1	0.857	—
<i>Phase 2 — Blind Self-Audit</i>		
Audit AUROC ↑	0.7171	[0.628, 0.816]
AUROC (model-planted control)	0.6973	—
Stylistic artifact Δ	+0.016	negligible
Planted detection rate	90%	—
<i>Phase 3 — Belief Revision</i>		
Hold rate ↑	100%	—
Revise rate ↑	50%	—
Sycophancy Index ↓	0.00	[0.000, 0.000]

↓ = lower is better; ↑ = higher is better.

7.2. Per-Category Diagnostic Profile

Table 4: Per-category accuracy and calibration. All $N = 25$ per category.

Category	Accuracy	Brier	95% CI	Interpretation
Logic	100%	0.000	[1.00, 1.00]	Perfect; CI degenerate
Fabricated facts	96%	0.036	[0.88, 1.00]	Strong; near-perfect recall
Linguistic puzzles	96%	0.040	[0.88, 1.00]	Strong rule induction
Arithmetic	92%	0.080	[0.80, 1.00]	Recovered after parser fix
Calibration traps	76%	0.240	[0.56, 0.92]	Moderate; intuition bias
Distorted facts	56%	0.436	[0.36, 0.72]	Primary weakness

The category profile reveals a clear performance gradient. Logic, fabricated facts, linguistic puzzles, and arithmetic sit above 90%—categories where the model’s parametric knowledge and systematic reasoning are sufficient. Calibration traps at 76% indicate residual intuition bias: the model resists the obvious wrong answer in most cases but not all. Distorted facts at 56% (95% CI [0.36, 0.72]) is the only category near chance performance and the only one where the CI does not exclude 50%, making it both the primary weakness and the most robustly characterised one.

7.3. The Formula Delta

The paper formula (Equation (6)) produces 0.8595 versus the canonical formula’s 0.7647—a gap of -0.095 . The driver is entirely Phase 3. The paper formula uses HoldRate (1.00) directly; the canonical formula uses $(\text{Hold} + \text{Revise})/2 = (1.00 + 0.50)/2 = 0.75$. This 0.25-point difference, weighted at $1/3 \approx 0.333$, contributes $0.333 \times 0.25 = 0.083$ to the gap. The remaining 0.012 comes from the weight difference on Phase 1 and Phase 2 (0.33 vs 0.25 and 0.33 vs 0.40 respectively).

Formula note

The paper formula would classify this model as near-Elite (0.8595 vs threshold 0.85), while the canonical formula correctly classifies it as Metacognitively Aware (0.7647). The 0.095-point gap is large enough to change tier classification and to affect comparative rankings across models.

7.4. Phase 2: Self-Audit Quality and Stylistic Confound

Audit AUROC of 0.717 (95% CI [0.628, 0.816]) places the model solidly in the Metacognitively Aware tier. The control condition (model-generated planted items, AUROC = 0.697) yields a stylistic artifact of $\Delta = +0.016$, which falls below the negligible threshold

(< 0.05). The 0.717 AUROC therefore reflects genuine discriminative self-awareness: the model has an internal correctness signal that allows it to distinguish, at above-chance rates, which of its Phase 1 outputs are likely correct versus incorrect.

Planted detection rate of 90% (18 of 20 planted items correctly flagged) is a complementary measure, though it depends on the rating threshold and is not as clean a metric as AUROC.

7.5. Phase 3: Sycophancy Resistance and Belief Revision

The Sycophancy Index of 0.00 with a 95% CI of [0.000, 0.000] is the strongest single finding: the model maintained every correct answer under invalid authoritative pressure. This was stable across independent Phase 3 runs. The zero CI is a consequence of all 10 sophistic challenges being met with MAINTAIN decisions.

The Revise Rate of 50% presents a more nuanced picture. With valid reasoning-rich corrections, the model updated 5 of its 10 wrong beliefs. The 5 that were not corrected represent a genuine residue of posterior rigidity—cases where the model received valid evidence but maintained its (wrong) prior. Notably, distorted-fact questions are likely over-represented in these non-revised cases, given the 56% Phase 1 accuracy on that category: wrong distorted-fact answers arise from the model accepting a false premise, and correcting them requires overriding a confident wrong belief.

7.6. Temperature Sensitivity Analysis

Table 5: Phase 3 metrics across temperature values (N=25 per category, $n = 20$ challenges per temperature).

Temperature	Hold rate	Revise rate	SI
0.0	90%	40%	0.10
0.3	90%	20%	0.10
0.7	80%	20%	0.20
1.0	100%	10%	0.00
SI range		0.00–0.20 (spread = 0.20)	

The SI range of 0.20 meets the moderate sensitivity threshold, meaning the Sycophancy Index should be treated as a (model, temperature) pair. The main evaluation at T=0.7 gives SI=0.20 in the sweep but SI=0.00 in the primary run, reflecting the stochasticity of the small Phase 3 sample (10 correct answers). Temperature has a non-monotonic effect on Revise Rate: T=0.7 produces the highest (40%) in the sweep, while T=1.0 produces the lowest (10%), contrary to the intuition that higher temperature promotes updating.

This non-monotonicity is likely an artefact of $n = 10$ per condition and should not be over-interpreted.

7.7. Domain-Weighted Composite Scores

Table 6: Domain-weighted composite scores using presets from Table 2.

Domain	Score	Calibration	AUROC	Belief
Medical	0.7941	0.50	0.35	0.15
Research	0.7686	0.30	0.45	0.25
General	0.7647	0.25	0.40	0.35
Legal	0.7640	0.20	0.25	0.55

Medical scores highest (0.7941) because it up-weights calibration (0.50), where the model performs well (Brier = 0.139). Legal scores lowest (0.7640) despite being numerically close: it up-weights belief revision (0.55), where the 50% Revise Rate creates a Phase 3 component of $(1.00 + 0.50)/2 = 0.75$ —reasonable, but the lowest among all components. For a legal deployment context, this means the model correctly holds firm under invalid pressure but fails to update half of its wrong positions when shown valid evidence—a meaningful liability in adversarial legal reasoning.

8. Discussion

8.1. Distorted Facts as the Metacognitive Frontier

The 56% accuracy on distorted facts (95% CI [0.36, 0.72]) is not an artefact of N ; it is the only category where the CI excludes neither 50% nor 70%, placing performance unambiguously between chance and competence. The mechanism is revealing. Distorted-fact questions embed a factual error in the premise and ask the model to engage with the question—correct performance requires flagging the premise error, not answering it as posed. This task sits at the intersection of three metacognitive capacities:

1. **World knowledge retrieval:** recognising that the premise contradicts stored knowledge.
2. **Epistemic monitoring:** detecting that the question as posed is unanswerable.
3. **Premise override:** actively preferring stored knowledge over the provided context, rather than deferring to the question’s framing.

Failure on distorted facts is therefore a form of pre-sycophancy: accepting a false premise before any social pressure is applied. A model that cannot maintain epistemic vigilance at the question stage is unlikely to maintain it under adversarial challenge. The Phase 3 Revise Rate of 50%—specifically the 50% non-correction rate—is plausibly driven by distorted-fact wrong answers that are also resistant to valid correction, precisely because the model has already accepted the distorted premise as ground truth.

8.2. Sycophancy Resistance vs. Belief Revision: A Dissociation

The co-occurrence of $SI=0.00$ (perfect sycophancy resistance) and $Revise\ Rate=50\%$ (partial belief revision) reveals an asymmetry in the model’s metacognitive architecture that is obscured by the paper formula’s exclusive focus on $HoldRate$. The model is highly resistant to being pushed off correct beliefs, but inconsistently receptive to being pulled toward correct beliefs from wrong starting positions.

This asymmetry has a natural explanation: the model’s internal confidence signal is well- calibrated on the correct answers challenged in Phase 3 (by selection, these are high- confidence correct answers), making them resistant to external pressure. The wrong answers challenged are lower-confidence by construction, and the model may be appropriately uncertain about whether the critic’s correction is itself valid—particularly for distorted facts, where the correction contradicts a premise the model had already accepted.

This dissociation is not captured by any single scalar metric, and its detection required the EPISTEMIC AUDIT separation of $Hold\ Rate$ and $Revise\ Rate$. Future benchmarks measuring sycophancy should report both.

8.3. The Calibration-AUROC Relationship

$Brier = 0.139$ and $AUROC = 0.717$ together paint a consistent picture: the model is reasonably calibrated at the aggregate level but has imperfect discriminative self-awareness at the item level. The aggregate calibration (accuracy $\approx 86\%$, mean confidence $\approx 86\%$ implied by $Brier = 0.139$ if confidence and accuracy co-vary appropriately) is largely maintained, but the $ECE = 0.138$ indicates that confidence and accuracy do not perfectly align within confidence bins.

The flat ECE across all bin counts ($M = 5, 10, 15, 20$ all yield 0.138) is a methodologically important result: it confirms that the calibration estimate is not sensitive to the arbitrary choice of M , a common but under-reported source of ECE instability.

8.4. Implications for Formula Standardisation

The 0.095-point gap between the canonical and paper formulas is large enough to change tier classification and would affect multi-model rankings if different models benefited

differently from the formula choice. We recommend that benchmarking publications adopting the EPISTEMIC AUDIT framework report:

1. The exact formula weights (α, β, γ) .
2. Both the canonical and paper scores if both are computed.
3. The Revise Rate separately from the Hold Rate.
4. The temperature used for Phase 3.
5. The N per category with bootstrap CIs.

Without these five disclosures, EPISTEMIC AUDIT scores from different papers are not comparable.

8.5. Limitations

Single-model evaluation. Results are reported for Gemini 2.5 Flash only. The benchmark’s discriminatory power across models—its ability to separate metacognitive profiles meaningfully—has not been demonstrated in this version. Future work should include DeepSeek R1, Llama 3.3 70B, and at least one further model for comparative analysis.

Temperature sensitivity. The SI range of 0.20 indicates moderate temperature dependence. At $N=10$ per Phase 3 condition, the non-monotonic behaviour of the Revise Rate across temperatures is likely stochastic noise rather than a genuine property of the model. Increasing Phase 3 to $n \geq 20$ per condition (i.e., challenging 20 correct and 20 incorrect answers) would substantially improve the statistical reliability of temperature sensitivity estimates.

Distorted-fact pool size. The 25 distorted-fact questions in this run draw from a pool of approximately 25 templates. At $N = 25$, each template is used approximately once, meaning the difficulty distribution is fully explored but there is no within-template sampling variance. Expanding the pool to 50+ templates would allow $N=25$ to be drawn with replacement and provide a more representative sample of the difficulty distribution.

Human baseline. No matched human baseline is available. The performance level labels (Metacognitively Blind, Partially Calibrated, Aware, Elite) are defined with reference to AUROC thresholds chosen on theoretical grounds, not empirical human norms. A future version of the benchmark should include a human-baseline calibration study using the same procedurally generated questions.

8.6. Future Directions

1. **Multi-model comparison at N=25:** running EPISTEMIC AUDIT against DeepSeek R1, Llama 3.3 70B, Qwen 2.5 72B, and Gemma 3 27B with identical methodology and N to produce the comparative table the paper requires.
2. **Cross-model Phase 2 audit:** having Model B audit Model A’s outputs to determine whether the AUROC deficit reflects self-knowledge specifically or general error discrimination.
3. **Multi-turn belief erosion:** extending Phase 3 to 3 rounds of escalating pressure to distinguish brittle from robust sycophancy resistance.
4. **Distorted-fact pool expansion:** growing to 50+ templates and testing whether distorted-fact performance improves with targeted fine-tuning on premise-verification tasks.
5. **Process-based reward alignment:** implementing the Brier-penalised reward function $R_{\text{meta}} = R_{\text{correctness}} - \lambda \cdot B$ in an open-source training pipeline to assess whether metacognitive benchmarking can be used as a training objective.

9. Conclusion

The EPISTEMIC AUDIT presents a methodologically corrected and statistically rigorous evaluation of metacognitive competence in large language models. Three measurement artefacts in the original implementation—a regex parser that truncated multi-line arithmetic answers, counterarguments too thin to constitute genuine epistemic challenge, and abstention signals that fired on correct distorted answers—were identified, corrected, and locked in with regression tests.

Evaluating Gemini 2.5 Flash at N=150, we find a model that is well-calibrated, genuinely self-aware (AUROC = 0.717, confirmed by stylistic confound control), and perfectly resistant to sycophancy under invalid pressure (SI = 0.00). The primary remaining weakness is distorted-fact recognition (56%), representing the benchmark’s hardest metacognitive challenge: overriding a plausible-sounding false premise with stored world knowledge, without any external confirmation. The 50% Revise Rate reveals a complementary gap: the model updates only half of its wrong beliefs when shown valid evidence, suggesting that its posterior-update mechanism is not symmetric with its prior-maintenance mechanism.

The formula delta of -0.095 documents a real methodological stakes: the paper formula inflates the composite score enough to change tier classification. Formula transparency is not a minor technicality but a prerequisite for cross-paper comparability in

any benchmarking field, and the EPISTEMIC AUDIT community should adopt the canonical formula (Equation (5)) as the standard.

As LLMs are deployed in consequential domains, the ability to know what one does not know becomes as important as the ability to answer correctly. Epistemic Audit provides a structured, reproducible, and statistically grounded instrument for measuring this capacity.

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